# Latency-aware cross-venue arbitrage

## Problem & theme

Problem 6, Cross-Asset Statistical Arbitrage with Latency Budgets (Theme 3). A correct signal is worthless if the gap closes before the order lands, so we price each opportunity at the latency it will face, then act, think longer, or skip. It suits us: we already record three venues and replay every fill.

## Understanding of the challenge

Gaps decay while we compute and orders travel, so we rank candidates by the edge left after latency and fees.

- **Venue offset.** Binance and Bybit sit a drifting 0.4–3 bps apart; most raw gaps are that offset, repaid when positions close.
- **Sampling artefacts.** About half the Bybit spot–perpetual gaps above 1 bps in the 100 ms archive are absent from 5.8 hours of our 10 ms recordings.

## Proposed approach

Each candidate i is scored in bps at latency τ (feed delay, compute, order travel):

$$S_i(\tau) \;=\; q_i(\tau)\,(\hat{e}_i - c_i) \;+\; (1 - q_i(\tau))\,\bar{\ell}(\tau)$$

q is the joint probability that both IOC legs fill at the observed prices after τ, learned from replayed fills; ê is the gap against the usual offset, c both legs' fees, $\bar{\ell}$ the average outcome of incomplete fills.

- **Data (about 260 GB raw).** Bybit 100 ms books, validated against trades, and Binance ticks (May 2023–Jan 2024; Bybit also Sep–Oct 2026); Tardis.dev ticks on Binance, Bybit, OKX (one day a month, 2020–2026); funding; our 10 ms recordings of ten coins on three venues.
- **Fast path and prior.** Fee-adjusted gap against a trailing 5-minute offset, then a calibrated six-feature logistic model of q (1.5 µs).
- **Slow model.** Gradient boosting on 27–43 causal features (Kalman, Ornstein–Uhlenbeck); the 23-feature live path takes about 30 µs.
- **Replay.** Order latency 50 ms Binance, 8 ms Bybit (estimated; no real orders), Bybit feed delay 45 ms, top-tier fees, $10k trades, $50k cap, optimistic and conservative fills.
- **Cross-asset pairs** (spot vs perpetual, ETH/BTC): decay AUC 0.78–0.91 against 0.49–0.87 for gap size; no edge after fees.

## Minimum viable (built)

| Problem bullet | What exists | Demo |
|---|---|---|
| Detect mispricings with a threshold or statistical model | Offset-aware fast path; Kalman/OU fair value | Paper trader logs every decision |
| Report how fast opportunities decay | On our recordings, 61% of raw BTC Binance–Bybit gaps above 1 bps survive 50 ms, 31% 500 ms | Dashboard |

## Stretch ambitions (both built)

- **Simulated end-to-end latency budget.** Next step from decay: P&L at our latencies, model time included.
- **Paper-trading demo.** Live feeds, simulated fills, P&L; fast path and raw-gap prior so far.

## Novelty / differentiation

The starter baseline thresholds a paired price series. Two departures each changed the result (development days):

- **Gaps against the usual offset, with honest inventory.** Raw-gap "profit" was mostly the offset and vanished once each trade carried its position, funding and depth-limited close.
- **Decay-gated decisions.** Trading every gap loses $94–925 a day per coin. Gated, with optimistic fills, the cascade profits on five of nine coins (intervals above zero; $15 a day on BTC and ETH, BTC 95% CI $7 to $23, 205 days) and beats a gap-size-only model on ETH, SOL, WLD and XRP (BTC borderline). Conservative fills break even on the

100 ms archive; on Tardis, which sees every quote, they keep $10–12 a day on BTC, ETH and SOL, every interval above zero. OKX pairs show no edge at assumed OKX costs.

The cascade calls the slow model only when the prior is unsure; its P&L stays within $7 a day of "slow always".

## Evaluation plan

Five policies, identical replays: naive, gap-size-only model, prior, slow always, cascade.

| Metric | Compared against |
|---|---|
| Net P&L/day after all costs at 1–50 ms latency; paired day-bootstrap 95% CIs | Naive, gap-size rule |
| AUC, Brier skill, calibration of q and of P(gap open after τ) | Gap size; constant survival rate |
| Median stage latency (µs); p99 on the VM | Order-latency budget |

Walk-forward folds, one-day embargo, training-day thresholds. BTC spot–perpetual AUC at 500 ms falls from 0.78 on the archive to 0.65 on our recordings, so archive AUCs look optimistic. Two locked test sets (Bybit 10–18 Oct 2026; Binance vs Bybit Feb–Mar 2024, setup fixed before download) are opened once, results reported in full.

## Feasibility and risks

### Team

- **Mohnish Rawat:** data and systems: recorders, Parquet pipeline, replay and inventory engine (Python, Numba).
- **Anay Sharma:** statistics: offset-aware evaluation, latency and fees, live slow path, dashboard.
- **Arnav Jain:** backtesting and front end.

MVP and both stretch goals run (165 tests, including a lookahead check).

### Milestones

| Week (from 10 Nov) | Owner | Milestone | Done when |
|---|---|---|---|
| 1–2 | Mohnish | 10 ms recording from Singapore, Tokyo VMs | Two weeks, under 1% missing |
| 1–2 | Anay | Offset-aware prior, slow model live | Cascade runs live |
| 3 | Anay | Interval-censored survival model | Beats 0.74 AUC at 50 ms |
| 3 | Arnav | p50/p99 stage latency on the VM | On the dashboard |
| 4 | Anay, Mohnish | Maker-taker with queue cancellations | Re-scored; Midway report 1 Dec |
| 5 | All | Open locked tests once; slides | Reported in full |

### Risks

- **Fees exceed most gaps.** Top-tier taker fees are 1.7–3 bps a leg; 2026 gaps rarely pass 2 bps. We report conservative fills; maker-taker replays $17–19 a day on 11 held-out January 2024 days.
- **Data gaps.** Outages are found from silences and skipped sequence numbers; their episodes are kept censored (lifetime at least what was observed).

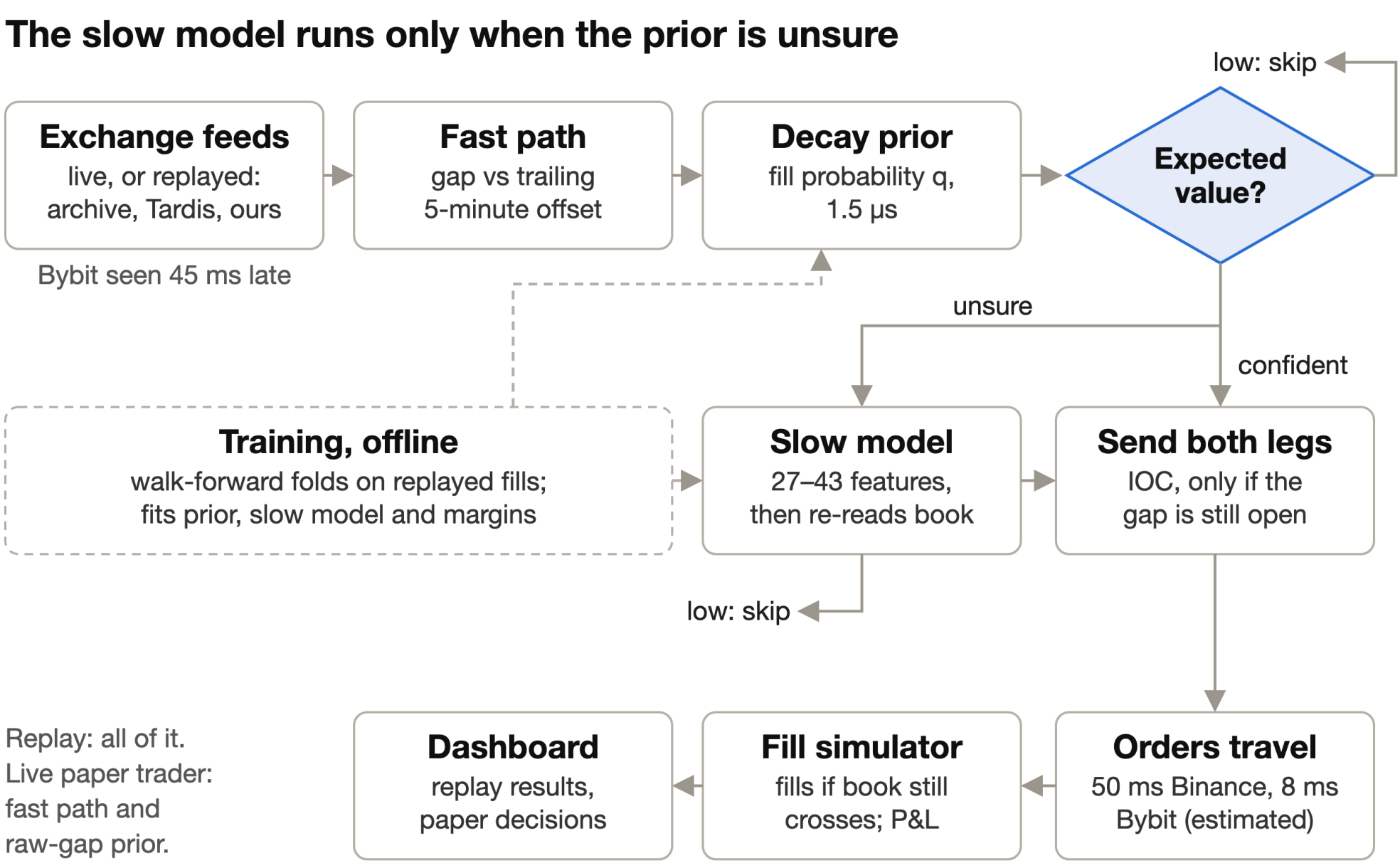



**Figure 1.** Decision pipeline. Timings measured on our machine.